DLS Buddy — Quickstart Guide

Applies to v1.21.0 and later

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1. What You Will Do and See

This guide walks you through one complete analysis on a dataset that ships with the program: **PEG 300k in water**, the synthetic Clean set in `test-data/Synthetic/Clean/`. You will load two files, confirm a handful of parameters, run one DLS fit and one SLS fit, export a result, and set the program to reopen your session next time. At the end you will have on screen:

- a **DLS** result — a hydrodynamic radius R_h of about 17.6 nm at one angle;
- an **SLS** result — a Zimm plot giving the weight-average molecular weight $M_w \approx 3.0 \times 10^5$ g/mol, the radius of gyration $R_g \approx 27$ nm and the second virial coefficient A_2.

Every step names the control to use and what you see when it has worked. Follow the steps in order; the values to type are given as you need them. The dataset's own `parameters.txt` lists the same values and the true sizes the set was generated with, so you can check any result against it. Once you have done it once, the same path — **load, confirm, Run, export** — is the path for your own data.

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2. Launching the Program

DLS Buddy needs **Python 3.13** installed; the README's *Running it* section says where to get it. Then double-click the launcher for your operating system in the program folder:

- **Windows** — `Launch DLS Buddy (Windows).bat`
- **macOS** — `Launch DLS Buddy (MacOS).command` (on the first run, macOS Gatekeeper may refuse an unsigned script — right-click it → **Open** → confirm once, and it opens normally thereafter)

The first launch sets up its own environment and installs the packages it needs; a console window reports "Setting up environment for the first time, this may take a minute…". Later launches open the program directly.

The window is a **Workspace** sidebar on the left, which will list your loaded data, and six tabs across the top: **Data**, **DLS**, **SLS**, **Cross-Sample**, **Utilities** and **Settings**. You will use the first three, then Settings once at the end.

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3. Loading the Example

1. **Click Load DLS correlogram…** and open `DLS - PEG 300k - 0.6 mg per mL.ASC` from `test-data/Synthetic/Clean/`. The sidebar shows **(unconfirmed sample: 298.15 K)** with eight measurements under it, one per scattering angle from 35° to 140°. Under each measurement sits the count-rate trace the file recorded, labeled with the file's own description line, "PEG 300k in Water, c = 0.6 mg/mL".
2. **Click Load SLS intensities…** and open `SLS Zimm - PEG 300k.csv` from the same folder. A second entry, **(unconfirmed sample: water)**, appears with six measurements: the solvent blank at 0 mg/mL and five concentrations from 0.2 to 1.4 mg/mL.

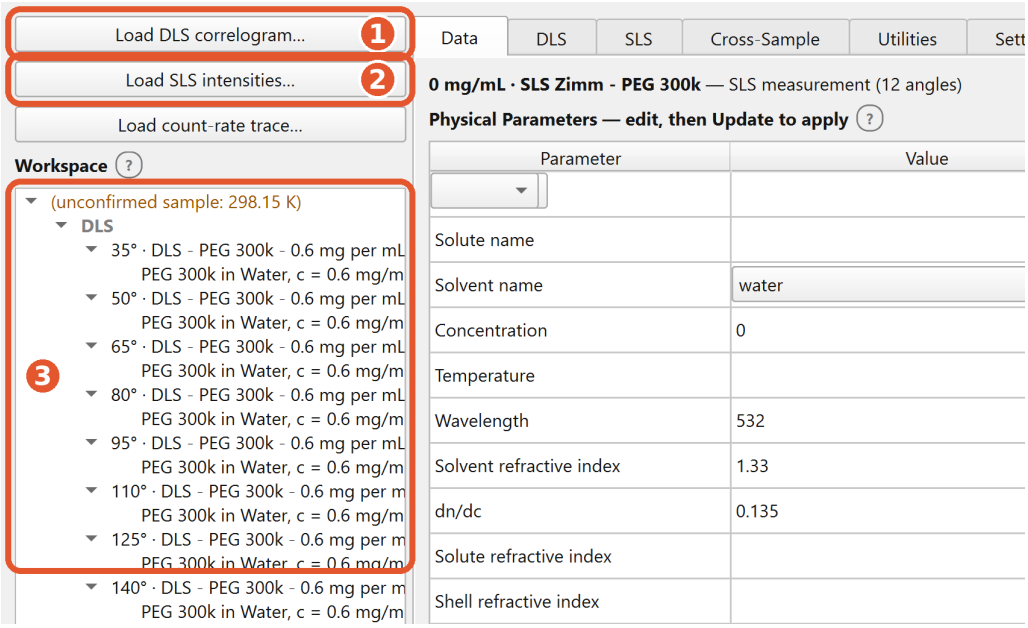


Figure 1: Loading the example: **Load DLS correlogram…** (1) and **Load SLS intensities…** (2). The DLS file's eight angles, each with its trace, appear in the sidebar (3) under an *unconfirmed* sample; the SLS file's entry follows below the crop.

Both entries are "unconfirmed" because a **sample** is keyed by the solute name, solvent and temperature in each measurement's **Physical Parameters** — never by a file's description line — and the DLS measurements lack the solute name while the SLS ones lack the temperature. The next chapter supplies both, and the two entries join into one sample.

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4. Confirming the Parameters

The **Data** tab shows the **Physical Parameters** of the measurement selected in the sidebar. The files have pre-filled most of them; you supply the rest and press **Update** to commit. A value entered here reaches every measurement of the same sample, so within a sample you type it once.

1. **Click the first DLS measurement** (`35° · DLS - PEG 300k - 0.6 mg per mL`) in the sidebar. On the **Data** tab, type `PEG 300k` in **Solute name**. Press **Enter** to move to the next field.
2. **Pick `water` in Solvent name.** The dropdown lists the library's solvents in two sections; `water` is under the **Measured properties** heading. The program fills **Solvent refractive index** (1.3348) and **Viscosity** (0.891 mPa·s) from its solvent library at the file's temperature and wavelength, and marks each with a small **teal dot** — the dot means *proposed by the program*. Leave **Concentration** blank; the DLS fit in the next chapter does not need it.
3. **Press Update.** The sidebar entry becomes **PEG 300k / water @ 298.15 K**.
4. **Click the first SLS measurement** (`SLS Zimm - PEG 300k`, the solvent blank at 0 mg/mL). Its measurements are still a sample of their own, so type `PEG 300k` in **Solute name** here too, and `25` in **Temperature** (the unit dropdown reads °C). The file already carries the solvent, the wavelength, each concentration and **dn/dc** = 0.135 mL/g, which M_w depends on. With the temperature set, the refractive index is proposed from the library as before.
5. **Press Update.** The six SLS measurements join the DLS sample, and the sidebar shows one entry, **PEG 300k / water @ 298.15 K**, with both files' measurements under it.

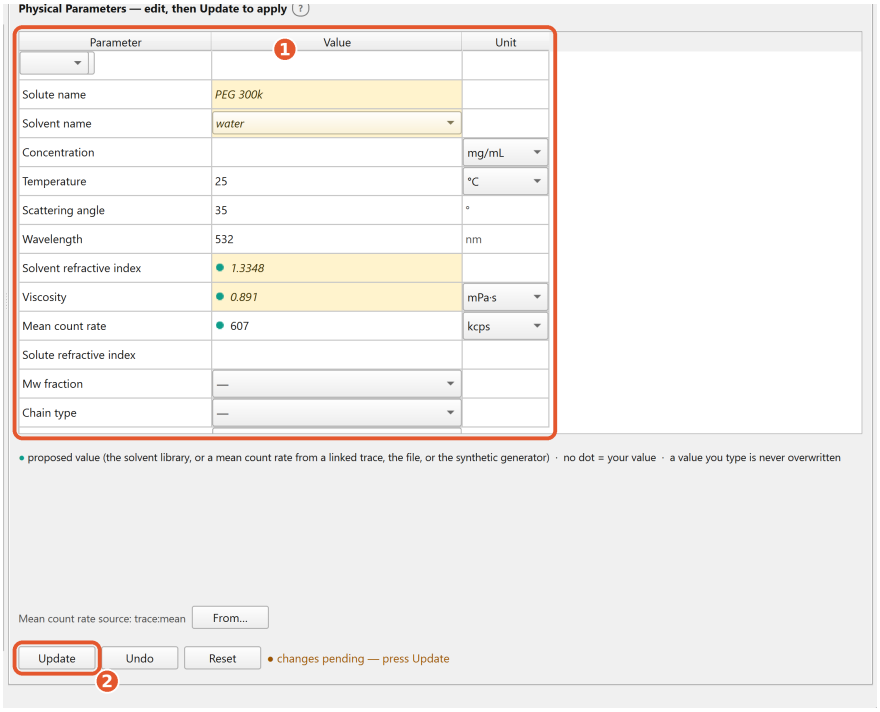


Figure 2: The **Data** tab with the DLS measurement's parameters (1) as they stand before **Update** (2): the typed solute name, the picked solvent, and the teal-dotted values the program proposed.

5. The DLS Result

1. **Open the DLS tab.** It opens on the **Correlogram** sub-tab.
2. **Tick** `35° · DLS - PEG 300k - 0.6 mg per mL` in the **Measurements to Plot** list at the top left. The raw correlogram, $g_2(\tau)-1$ against delay time, appears in the plot.
3. **Check the fit settings** under **Parametric Fit**: **Method:** `Cumulant` and **Cumulant order:** `2`, the defaults. Leave the delay window and baseline as proposed.
4. **Press Run.** The fitted curve is drawn over the data, the residuals appear beneath it, and the **Cumulant results — apparent sizes** table lists the measurement with R_h = **17.6 nm** and a polydispersity index PDI = **0.0144**, with a ✔ beside the PDI. The **Details** table below repeats them beside the decay rate and the fitted moment.

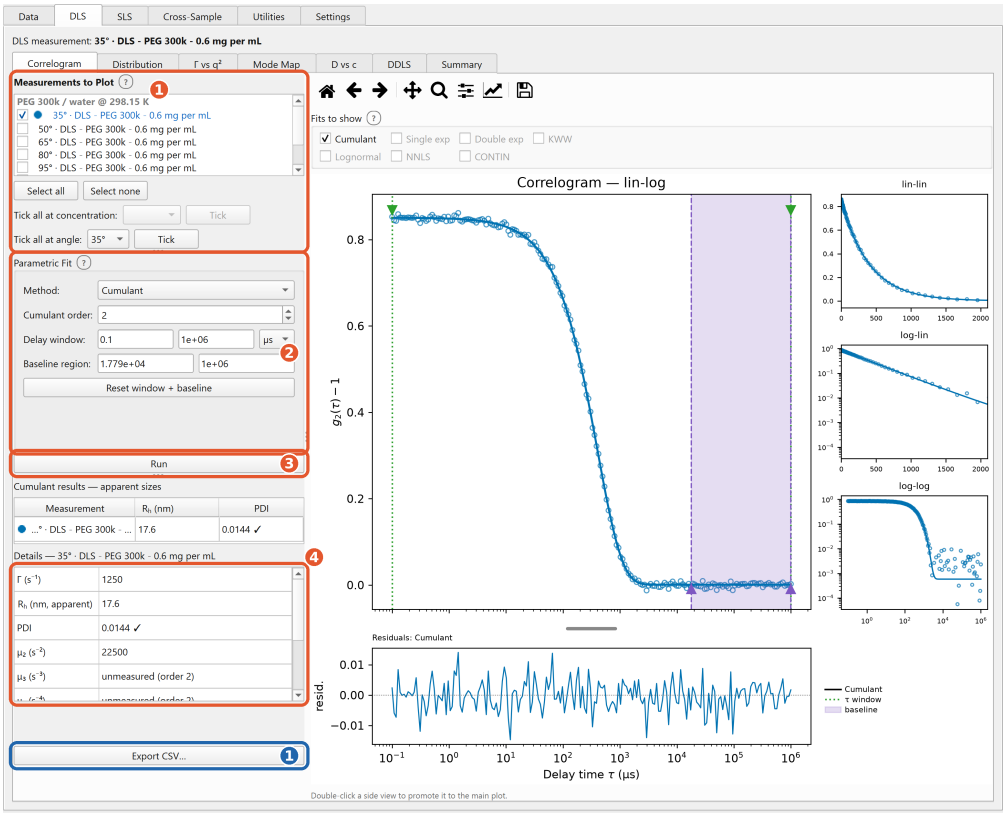


Figure 3: The **DLS** tab after **Run**: the ticked measurement (1), the fit settings (2), **Run** (3) and the result — the **Cumulant results** table and the **Details** beneath it (4). The correlogram, its cumulant fit and the residuals fill the plot. *Blue:* **Export CSV...** writes the result on screen.

This R_h is an **apparent** size: it comes from one angle at one finite concentration, which is what a single DLS measurement can give, and it sits a little below the 18.0 nm the dataset was generated with (`parameters.txt`) for that reason. The ✔ says the fit's preconditions held — the PDI is within the range the cumulant method describes.

6. The SLS Result

1. **Open the SLS tab.** It opens on the **Analysis** sub-tab, and **SLS Sample** at the top already shows `PEG 300k / water @ 298.15 K` — the whole set, the solvent blank and the five concentrations, is analyzed together.

2. **Type `100000` in Calibrant intensity** in the **Calibration** box — the toluene standard's intensity this dataset was recorded against. The box's other fields already hold the dataset's values, and **Standard refractive index** is proposed by the program (teal dot): accept it. Press the box's own **Update**. The line **Calibration $\mathbf{k_c}$** then reads `committed = 3.1496e-10`: the constant the program computed from that one calibrant point.

3. **Check Method:** `Zimm`, the default, in the **Analysis** box.

4. **Press Run.** The **Zimm plot** appears — one series per concentration, the two dashed extrapolations to zero concentration and zero angle, and the common intercept — and the **Results** table beneath it reads:

Method	Zimm
M_w (g/mol)	(2.992 ± 0.007)e5
R_g (nm)	26.93 ± 0.18
A_2 (mol·mL/g^2)	(1.24 ± 0.04)e-4
R^2	0.991

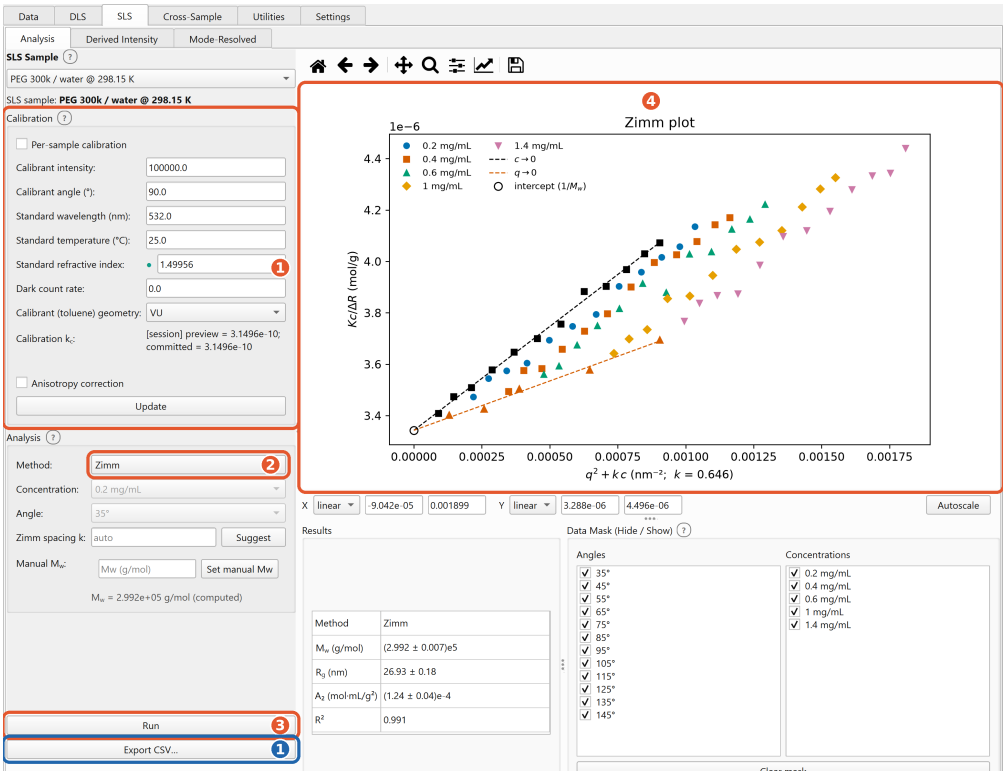


Figure 4: The **SLS** tab after **Run**: the calibration box (1), **Method:** (2), **Run** (3) and the Zimm plot (4), with the **Results** table beneath it. *Blue:* **Export CSV...** writes the result.

These are **thermodynamic** values — extrapolated to zero angle and zero concentration — and they recover the values the dataset was generated with (M_w 3.0 × 10^5 g/mol, R_g 27 nm, A_2 1.2 × 10^{-4} mol·mL/g^2) to within about one standard error.

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7. Exporting Results and Restoring the Session

1. **Press Export CSV...** at the bottom of the SLS tab's control column. The save dialog proposes `PEG 300k-water_zimm.csv`, named after the sample and the analysis. Save it. The file opens directly in OriginLab, with the axis names and units already in place.
2. **Open the Settings tab, tick Reopen last session on startup** — the **Session** row, the last row under the **Appearance** heading at the bottom of the form — **and press the tab's Update.** The setting takes effect on that **Update**. After it, the program saves your session when you close it and brings it back — data, parameters and results — the next time you launch.

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8. Where to Go Next

That is the whole path, and it is the same for your own data: load, confirm, **Run**, export. The **User-Manual** covers every supported file format and what each one pre-fills, every parameter and unit, every method on the DLS and SLS tabs, and the Cross-Sample and Utilities tools; the README has a terminal recipe for the setup the launcher performs. The **Theory-and-Equations-Guide** gives the equations and the literature behind each method, including how the cumulant R_h and the Zimm extrapolation are defined.